

Microsoft Security Baselines



Windows products out of the box

- Often focused more on availability and delivery instead of security
- Very generic in nature to fit in automatically in most of the places
- MVP in terms of security

Why baselines

- Windows 10 contains over 3000 policy settings
 - In addition over 1800 internet explorer 11 settings
- Security part is only small fraction and often overseen due to the need to focus on availability
- The baselines are designed for well-managed, security-conscious organizations in which standard end users don't have administrative rights
- A baseline enforces a setting only if it mitigates a contemporary security threat and doesn't cause operational issues that are worse than the risks they mitigate.

Getting the baselines

- Security baselines are part of Microsoft Security Compliance Toolkit and downloadable from internet
- <https://www.microsoft.com/en-us/download/details.aspx?id=55319>

Important! Selecting a language below will dynamically change the complete page content to that language.

Select language

English ▾

Download

Expand all | [Collapse all](#)

Details

Version:	Date Published:	File Name:	File Size:
1.0	10/31/2023	Windows 10 Version 1809 and Windows Server 2019 Security Baseline.zip	1.3 MB
			903.4 KB
		Windows 10 Version 1507 Security Baseline.zip	1.5 MB
		Windows 10 Version 1607 and Windows Server 2016 Security Baseline.zip	519.2 KB
			1.5 MB
		LGPO.zip	313.9 KB
		PolicyAnalyzer.zip	1.5 MB
		SetObjectSecurity.zip	452.4 KB
		Windows 10 Version 20H2 and Windows Server Version 20H2 Security Baseline.zip	1.3 MB
		Windows 10 Update Baseline.zip	1.2 MB
		Windows Server 2022 Security Baseline.zip	1.2 MB
		Windows 11 Security Baseline.zip	1.4 MB
		Windows 10 version 21H2 Security Baseline.zip	1.2 MB
			689.2 KB
		Windows 11 version 22H2 Security Baseline.zip	338.6 KB
			1.2 MB
		Windows 10 version 22H2 Security Baseline.zip	
		Microsoft 365 Apps for Enterprise 2306.zip	
		Microsoft Edge v117 Security Baseline.zip	
		Windows 11 v23H2 Security Baseline.zip	

Important takeaways

- When applying the policy in local group policy (non domain joined computer) then LGPO executable is needed
- Make sure you apply the correct baseline to correct target operating system (version)
- During testing applying Windows 10 baseline to windows 7 host caused it to randomly get BSOD

PolicyAnalyzer

- Policy Analyzer is a lightweight utility for analyzing and comparing sets of Group Policy Objects (GPOs)
- Policy analyzer works with .PolicyRules xml files which describe the policy settings
- PolicyRules files can be applied with LGPO.exe v3 tool
- Has tool to convert GPO backups into PolicyRules file

```
C:\Users\Administrator\Downloads\PolicyAnalyzer\PolicyAnalyzer_40>GP02PolicyRules.exe ../../rules .\OurRules.PolicyRules

GP02PolicyRules.exe - Convert GPO backup(s) to Policy Analyzer .PolicyRules file
Version 4.0.2004.13001
Copyright (C) 2009-2020 Microsoft Corporation
Security Compliance Toolkit - https://www.microsoft.com/download/details.aspx?id=55319

Building PolicyRules from the following GPOs:

    AllowIPSecTelnet
    Default Domain Controllers Policy
    Default Domain Policy
    DisableUpdates
    IPSec

GPO(s) written to ".\OurRules.PolicyRules".

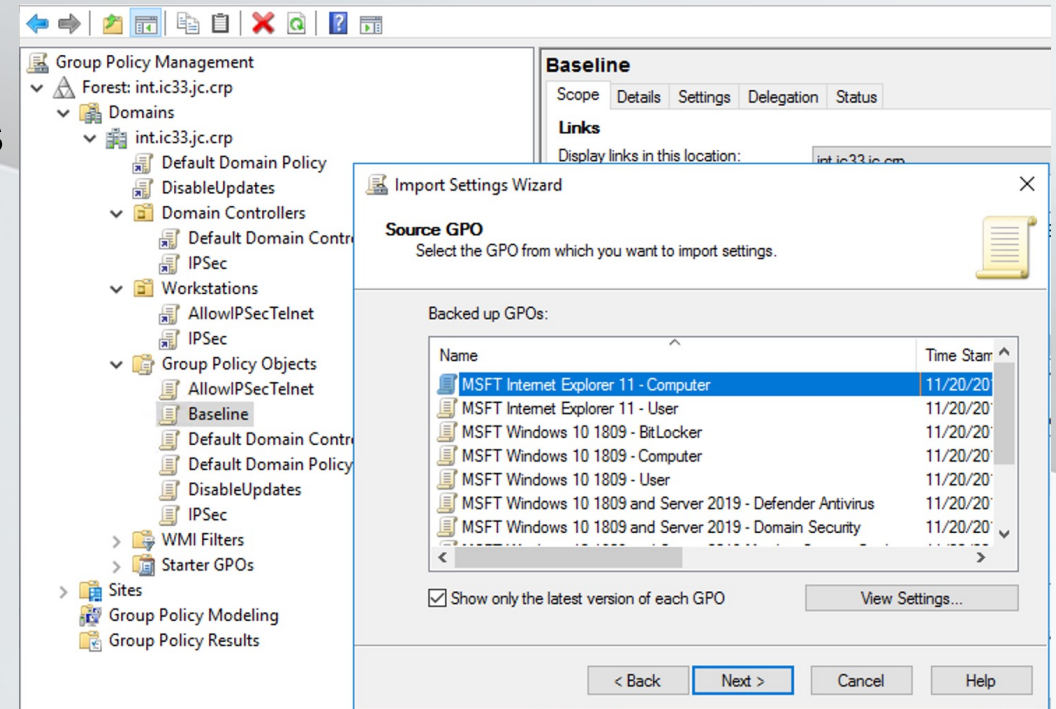
C:\Users\Administrator\Downloads\PolicyAnalyzer\PolicyAnalyzer_40>
```

PolicyAnalyzer to check IPsec rules

Policy Viewer - 56 items			
Clipboard View Export Options			
Policy Type	Policy Group or Registry Key	Policy Setting	OurRules
HKLM	Software\Policies\Microsoft\SystemCertificates\EFS	EFSBlob	01.00.01.00.01.0...
HKLM	Software\Policies\Microsoft\SystemCertificates\EFS\Certificates\6CD4E0FFC1A...	Blob	02.00.00.00.01.0...
HKLM	Software\Policies\Microsoft\SystemCertificates\EFS\CRLs		[[[create key]]]
HKLM	Software\Policies\Microsoft\SystemCertificates\EFS\CTLs		[[[create key]]]
HKLM	Software\Policies\Microsoft\Windows\WindowsUpdate\AU	NoAutoUpdate	1
HKLM	SOFTWARE\Policies\Microsoft\WindowsFirewall	PolicyVersion	538
HKLM	SOFTWARE\Policies\Microsoft\WindowsFirewall\ConSecRules	{89D3C4EF-044E-4427-B531-978...	Action=Boundary...
HKLM	SOFTWARE\Policies\Microsoft\WindowsFirewall\DomainProfile	DefaultInboundAction	1
HKLM	SOFTWARE\Policies\Microsoft\WindowsFirewall\DomainProfile	DefaultOutboundAction	0
HKLM	SOFTWARE\Policies\Microsoft\WindowsFirewall\DomainProfile	EnableFirewall	1
HKLM	SOFTWARE\Policies\Microsoft\WindowsFirewall\FirewallRules		[[[create key]]]
HKLM	SOFTWARE\Policies\Microsoft\WindowsFirewall\FirewallRules	{22CD58FB-1C97-48E8-9C4A-E49...	v2.26 Action=Allo...
HKLM	System\CurrentControlSet\Control\Lsa	NoLMHash	1
HKLM	System\CurrentControlSet\Services\LanManServer\Parameters	EnableSecuritySignature	1
HKLM	System\CurrentControlSet\Services\LanManServer\Parameters	RequireSecuritySignature	1
HKLM	System\CurrentControlSet\Services\Netlogon\Parameters	RequireSignOrSeal	1
HKLM	System\CurrentControlSet\Services\NTDS\Parameters	LDAPServerIntegrity	1
Security Template	Kerberos Policy	MaxClockSkew	5
Security Template	Kerberos Policy	MaxRenewAge	7
Security Template	Kerberos Policy	MaxServiceAge	600
Security Template	Kerberos Policy	MaxTicketAge	10
Security Template	Kerberos Policy	TicketValidateClient	1
Policy Path:			
Computer Configuration			
... \Windows Firewall with Advanced Security\... Rules			
{22CD58FB-1C97-48E8-9C4A-E498E4594DE9}			
OurRules:			
Option: Inbound: telnetIPSec			
v2.26			
Action = Allow			
Active = TRUE			
Dir = In			
Protocol = 6			
LPort = 23			
Name = telnetIPSec			
Security = AuthenticateEncrypt			
Data: v2.26 Action=Allow Active=TRUE Dir=In Protocol=6 LPort=23 Name=telnetIPSec Security=AuthenticateEncrypt			
Type: REG_SZ			
GPO: AllowIPSecTelnet			

Operating system specific baselines

- Packed file contains
 - Script to apply baseline policies locally
 - GPOs to apply via Group Policy
 - Multiple GPO backups for specific settings



Task 1

- Apply the correct **LOCAL** baseline policies on WS2

Task 2

- Apply correct baseline GPOs for dc1 (Server 2016)
 - Domain Controller
 - Domain Security
 - Credential Guard
 - Defender
 - Member Server Baseline - User
 - Member Server Baseline - Computer